

Yousra Al-Idrisi

B.A. Mathematics, UC Berkeley · M.A. Mathematics, UC Santa Cruz

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RESEARCH INTERESTS

Geometric group theory and low-dimensional topology. Interests include non-positive curvature, hyperbolic manifolds, Artin and Coxeter groups, random groups, and knot and link invariants.

EDUCATION

University of California, Santa Cruz
M.A. in Mathematics (Advisor: [Jiayin Pan](#))

Santa Cruz, California
August 2026

University of California, Berkeley
B.A. in Mathematics

Berkeley, California
August 2020

TEACHING

Private Tutoring

Tutor

Algebra I

Geometry

Algebra II / Trigonometry

Pre-Calculus

AP Calculus AB

AP Calculus BC

Multivariable Calculus

Differential Equations

Discrete Mathematics

Linear Algebra

February 2023 – Present

University of California, Santa Cruz

Teaching Assistant

MATH 23B – Vector Calculus

Summer 2026

MATH 19B – Calculus for Science, Engineering, and Mathematics

Spring 2026

MATH 16B – Mathematics for Life Sciences

Winter 2026

MATH 23A – Vector Calculus

Summer 2025

MATH 19B – Calculus for Science, Engineering, and Mathematics

Fall 2024, Winter 2025, & Spring 2025

MATH 21 – Linear Algebra

Summer 2023

MATH 11B – Calculus II with Applications

Spring 2023

MATH 11A – Calculus I with Applications

Winter 2023

MATH 3 – Precalculus

Fall 2022

University of California, Berkeley

Tutor

Math 3 – Precalculus

January 2018 – August 2018

Math 16A – Analytic Geometry and Calculus

Math 16B – Analytic Geometry and Calculus

Math 51 – Calculus I

Math 52 – Calculus II

Math 53 – Multivariable Calculus

Math 54 – Linear Algebra and Differential Equations

Math 55 – Discrete Mathematics

Math 56 – Linear Algebra

Southwestern College

Tutor

Math 60 – Intermediate Algebra I

August 2015 – June 2017

Math 72 – Intermediate Algebra I and II

Math 101 – College Algebra

Math 104 – Trigonometry

Math 244 – Precalculus with Trigonometry

Math 250 – Analytic Geometry and Calculus I

Math 251 – Analytic Geometry and Calculus II

Math 252 – Analytic Geometry and Calculus III

Math 253 – Intro to Differential Equations

Math 254 – Intro to Linear Algebra

RESEARCH EXPERIENCE

Master's Thesis: *Geometry of Nonpositive Curvature and the Flat Subspace Theorem* September 2025 – August 2026

University of California, Santa Cruz

- Gave a self-contained proof of the Gromoll–Wolf Flat Subspace Theorem, showing that solvable groups of semi-simple isometries acting on nonpositively curved spaces preserve a Euclidean flat on which the induced action is crystallographic
- Developed the underlying comparison geometry from first principles, including Rauch's comparison theorem, the Cartan–Hadamard theorem, and rigidity of geodesic triangles and quadrilaterals under curvature bounds

Advisor: [Jiayin Pan](#)

Independent Research in Knot Theory and Low-dimensional Topology

September 2024 – June 2025

University of California, Santa Cruz

- Investigated a conjecture relating the Wirtinger number of a link to the zero forcing number of a graph, considering both alternating and non-alternating knot diagrams

Advisor: [Ryan Blair](#)

Independent Study in Geometric Group Theory

Fall 2024

University of California, Santa Cruz

- Advanced independent study in geometric group theory, exploring hyperbolic groups, quasi-isometries, and Coxeter groups using *Office Hours with a Geometric Group Theorist* as a primary textbook

Advisor: [Nicolas Brody](#) & [Kasia Jankiewicz](#)

Direct Study Program

January 2018 – May 2018

University of California, Berkeley

- Gained first research experience in geometric group theory by studying foundational material on groups, spaces, and free groups using *Office Hours with a Geometric Group Theorist* as a primary textbook

Advisor: [Nicolas Brody](#)

PROFESSIONAL EXPERIENCE

Artificial Intelligence Scientist

February 2021 – October 2022

Naval Information Warfare Center Pacific, San Diego, CA

- Investigated topological data analysis and the consequences of Lipschitz regularization in machine learning classification tasks, with emphasis on geometric and topological structure of high-dimensional data

TALKS

The Inadequacy of Lipschitz Regularization in Machine Learning

Spring 2023

Graduate Colloquium, University of California, Santa Cruz

The Inadequacy of Lipschitz Regularization in Machine Learning

Spring 2023

Undergraduate Colloquium, University of California, Santa Cruz

AWARDS, GRANTS, AND FELLOWSHIPS

SMART Scholar, SMART Scholarship for Service Program, San Antonio, TX

August 2018

Two year stipend of \$25,000 with full tuition, fees, and health insurance support

Leadership Cal Alumni Award , University of California, Berkeley \$2,000 award for motivational leadership impacting academic, work, and community	August 2017
Honors Certificate Award , Southwestern College, Chula Vista, CA Recognition for completing a minimum of 15 units in Honors courses and maintaining a minimum GPA of 3.2	May 2017
Student of Distinction Award (SODA) , Southwestern College, Chula Vista, CA \$1,000 award for commitment to academic excellence and leadership	April 2016

CONFERENCES

International Conference on Machine Learning (ICML) , Baltimore, Maryland	July 2022
Graduate Research Opportunities for Women (GROW) , The University of Chicago	October 2021

SKILLS

Programming: CSS, HTML, JavaScript, LaTeX, MATLAB, Python, SageMath

MEMBERSHIPS

American Mathematical Society (AMS)
 Association for Women in Mathematics (AWM)
 Mathematical Association of America (MAA)
 National Association of Mathematicians (NAM)

REFERENCES

Available upon request.